Dragos Mihai Sorescu

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Honours B.Sc. in **Artificial Intelligence** Oshawa, Ontario, Canada

Specialized in **applied machine learning** & **reliable AI systems**

Education

Durham College, B.Sc. in Artificial Intelligence Sep 2023 – **present**

- **Honour Roll:** Fall 2025, Winter 2026
- **Relevant Courses:** Machine Learning, Computer Vision, Natural Language Processing, Statistics, Linear Algebra, Systems Analysis and Design, Computer Architecture

Experience

Research Assistant, Durham College AI Hub May 2026 – **present**

- Built the agentic core for a **multi-persona LLM chatbot** using **AWS Lambda** and **Bedrock**, supporting persona routing, multi-turn context, and SSM-managed prompts; load-tested to **1.8s p95 latency** via Locust
- Implemented **retrieval-augmented generation** (RAG) with **Amazon Bedrock Knowledge Bases** and **Amazon Titan Text Embeddings V2**, configuring document ingestion, chunking, and retrieval to ground chatbot responses in domain-specific source material
- Identified and fixed a **backend routing bug** that allowed rejected requests to invoke **LLM services**, saving **1.4K tokens** per affected request and reducing **cost-abuse risk**
- Re-architected backend routing with **TypeSafe's Jev decision model:** collapsed **3 sequential LLM classifiers into one parallel call**, cutting p50 latency **3.8s → 0.14s (~29x)** and tail 7.9s → 0.32s max (~25x) while holding **93.9% macro recall** across 6 intents
- Co-authored and presented a technical **advisory report** with a three-person team to the client, translating testing findings and **technical risks** into **actionable implementation recommendations**

Projects

Financial News Agent | Python, TypeScript, FastAPI, Claude API, Supabase, SvelteKit **Repository**

- Built **agentic financial assistant** with **Claude API** orchestrating price, news, financials, and insider-trade tools, paired with **SvelteKit frontend** featuring SSE streaming, live tool-call progress, and portfolio views
- Secured **FastAPI** endpoints with Supabase **JWT authentication**, row-level security, per-IP rate limiting, and input sanitization

Vision Transformer Research | Python, PyTorch, Scikit-learn, Pandas, Matplotlib, Seaborn **Repository**

- Benchmarked **8+ vision architectures**, including DINOv2, ViT, SwinV2, ResNet, and YOLO, for **binary skin cancer classification** using stratified 5-fold cross-validation
- Achieved **~92% sensitivity** at a clinically relevant threshold using ablation studies, freezing strategies, and ensemble evaluation
- Curated *MRA-MIDAS* from 3,000 to 906 **quality-controlled cases** and built preprocessing, augmentation, and training pipelines for **reproducible evaluation**

Temperature Modelling | Python, Pandas, Scikit-learn, XGBoost, FastAPI, Streamlit, GCP, Docker **Repository**

- Developed an **XGBoost** regression model for **temperature prediction** using historical weather data, applying cross-validation and regularization to reduce **overfitting** and **improve generalization**
- Achieved **R^2 0.991**, **MAE 0.71**, and **MSE 0.997** via 10-fold **RandomizedSearchCV** tuning depth, learning rate, and L1/L2 regularization
- Productionized the model as a **Dockerized FastAPI** service on **Google Cloud Run** and connected it to a Streamlit interface for on-demand predictions

Technologies

Languages & Tools: Python, C++, SQL, TypeScript, Git, Linux, Docker
ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, Pandas, NumPy, Matplotlib, Seaborn
LLMs & Testing: Claude API, Amazon Bedrock (Knowledge Bases), LangChain, Ragas, Pytest, Locust
Web & Cloud: FastAPI, SvelteKit, Supabase, Streamlit, AWS (EC2, S3, Lambda, API Gateway), GCP (Cloud Run)